

Farah Ben Amor

Engineering Student in Instrumentation & Industrial Maintenance

Embedded Systems · IoT · Power Electronics · Smart Energy Systems

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Seeking a Final-Year Internship (PFE) – 6 months – Starting February 2027

Profile

Engineering student specializing in embedded systems, IoT, and computer vision, with hands-on experience designing microcontroller firmware (STM32, ESP32), sensor-to-cloud IoT pipelines, and NPU-based embedded vision detection systems. Winner of multiple national hackathons, with strong involvement in student leadership and technological innovation.

Education

INSAT – National Institute of Applied Science and Technology, Tunis 2022 – 2027
Engineering Degree (Integrated Preparatory Cycle + Engineering Cycle) in Instrumentation & Industrial Maintenance

Experience

- Embedded Computer Vision Intern** | *MIGSens, Tunisia* Aug 2026 – present
- Designed an embedded vision pipeline on an STM32N6570-DK board (Neural-ART NPU) with an MB1280-3V3-C01 camera module: trained a MobileNetV2 multi-class classification model (fresh vs. rotten apples/bananas/oranges), quantized it to INT8, and deployed it for real-time fruit spoilage detection.
- Embedded Systems & IoT Intern** | *MIGSens, Tunisia* Jun – Aug 2026
- Designed an indoor/outdoor air-quality IoT monitoring station for the pharmaceutical industry end-to-end: researched and selected sensors, developed ESP32-S3 nodes through to a Node-RED + InfluxDB cloud pipeline, designed a real-time LVGL touchscreen interface, and wrote the user manual.
- Power Electronics & Renewable Energy Intern** | *SAGEMCOM, Tunisia* Jun – Jul 2025
- Sized and simulated a 5 kW solar VFD chain for a three-phase asynchronous pump: characterized the MPPT operating point, sized the IGBTs and DC-bus capacitor thermally and electrically, and produced the complete DC-AC single-line diagrams.

Projects & Achievements

- Embedded CAN Bus & IoT Vehicle Monitoring System** (PFA – Final Year Project) Feb – May 2026
- Designed a dual-STM32F407 CAN network simulating distributed ECUs, connected to the cloud via an ESP32 MQTT gateway with anomaly detection and a real-time Node-RED dashboard with alerting
- Remote Medical Monitoring System** (Real-Time Systems Project) Apr. – May 2026
- Developed non-blocking Arduino firmware for ECG/temperature acquisition and threshold-based clinical alert generation, with a real-time web dashboard; validated at 215 ms median latency and 96.8% detection rate.
- Hybrid Solar-Wind Renewable Energy Platform** (SunWhirl – 1st Place TSYP 13) Oct – Dec 2025
- Drove a dual-axis solar tracker via an Astral algorithm on a Raspberry Pi 5 and developed sensorless wind yaw control (P&O) on Arduino, with real-time I²C power acquisition at 300 ms via INA219
- AI-Powered Wearable Stress Monitor for Children with ASD** (CalmaWear – 1st Place TSYP 13) Oct – Dec 2025
- Led the development of the ESP32 firmware: multi-sensor acquisition (MAX30102, MPU6050, FSR402), real-time BPM/RMSSD/respiration computation, and Bluetooth Classic SPP streaming to a caregiver-facing mobile app (
- MPPT Optimization for PV Systems** (PPP – Personal and Professional Project) Feb – May 2025
- Benchmarked P&O, Incremental Conductance, Fuzzy Logic, and Neural Network MPPT algorithms under varying irradiance/temperature conditions to evaluate energy yield

Skills

Languages (Programming): C · C++ · Python

Microcontrollers: STM32 · STM32N6 (Neural-ART NPU) · ESP32/ESP32-S3 · Arduino · Raspberry Pi

Protocols: CAN · UART · SPI · I²C · MQTT · Bluetooth | **Development Tools:** Git

IoT & Cloud: Node-RED · InfluxDB · LVGL · Real-Time Dashboards

Embedded Vision & AI: TensorFlow · MobileNetV2 · ST Edge AI Core · INT8 Quantization

Simulation, EDA & CAD: MATLAB/Simulink · PSIM · PVsyst · LabVIEW · KiCad · Proteus · TIA Portal · SolidWorks · AutoCAD

Languages

French: Fluent | **English:** Professional – [TOEIC L&R 920/990](#) | **Arabic:** Native | **Turkish:** Intermediate

Awards & Certifications

- 1st Place – AI for Sustainability Hackathon (USMed), Nov 2024
- [Six Sigma Yellow Belt](#) (SSYB)
- [Certified SolidWorks Associate](#) (CSWA)

Leadership & Extracurricular Activities

- Jury Member** – H2GP Sprint Car Competition, Horizon Educational (*Switzerland*) June 29 – July 3, 2026
- Technical Workshop Facilitator** – IEEE Events (Summer School, Sparkle, NRTF) Sept.2023 – Sept.2026
- Vice-Chair of Technical Activities** – IEEE PES/PELS Joint Chapter, INSAT Aug. 2024 – Aug. 2025
- Project Manager** – National Re-Tech Fusion 1.0, IEEE Tunisia Dec. 2023 – June 2024